

DBMS - ORDER AND ORDER DETAILS MANAGEMENT

Project Title: Savana Online E-Commerce Database Management System

Skill Task : DBMS

1. Database Design

1.1 Orders Table

Attribute	Data Type	Constraint
Order_ID	NUMBER(5)	Primary Key
Customer_ID	NUMBER(5)	Foreign key
Order_Date	DATE	NOT NULL
Total_Amount	NUMBER(10,2)	NOT NULL
Order_Status	VARCHAR2(30)	—

1.2 Order_Details Table

Attribute	Data Type	Constraint
Order_Detail_ID	NUMBER(5)	Primary Key
Order_ID	NUMBER(5)	Foreign Key
Product_ID	NUMBER(5)	Foreign Key
Quantity	NUMBER(5)	NOT NULL
Unit_Price	NUMBER(10,2)	NOT NULL
Subtotal	NUMBER(10,2)	NOT NULL

2. Table Relationship

Customer → Orders

One Customer can place many Orders.

Orders → Order_Details

One Order can contain many Order_Details.

Product → Order_Details

One Product can appear in many Order_Details.

3. Table Creation

3.1 Create Order Table

```
SQL> CREATE TABLE SAVANA_ORDERS (  
 2     ORDER_ID NUMBER(5) PRIMARY KEY,  
 3     CUSTOMER_ID NUMBER(5),  
 4     ORDER_DATE DATE NOT NULL,  
 5     TOTAL_AMOUNT NUMBER(10,2) NOT NULL,  
 6     ORDER_STATUS VARCHAR2(30),  
 7     CONSTRAINT FK_ORDER_CUSTOMER  
 8     FOREIGN KEY (CUSTOMER_ID)  
 9     REFERENCES CUSTOMER(CUSTOMER_ID)  
10 );
```

Table created.

3.2 Create Order_Details

```

SQL> CREATE TABLE SAVANA_ORDER_DETAILS (
  2     ORDER_DETAIL_ID NUMBER(5) PRIMARY KEY,
  3     ORDER_ID NUMBER(5),
  4     PRODUCT_ID NUMBER(5),
  5     QUANTITY NUMBER(5) NOT NULL,
  6     UNIT_PRICE NUMBER(10,2) NOT NULL,
  7     SUBTOTAL NUMBER(10,2) NOT NULL,
  8     CONSTRAINT FK_DETAIL_ORDER
  9     FOREIGN KEY (ORDER_ID)
10     REFERENCES SAVANA_ORDERS(ORDER_ID),
11     CONSTRAINT FK_DETAIL_PRODUCT
12     FOREIGN KEY (PRODUCT_ID)
13     REFERENCES SHOP_PRODUCT(PRODUCT_ID)
14 );

```

Table created.

4.Order Data Insertion

```

SQL> INSERT INTO SAVANA_ORDERS
  2 VALUES
  3 (1001, 101, TO_DATE('21-09-2026','DD-MM-YYYY'), 25998, 'Confirmed');

1 row created.

SQL>
SQL> INSERT INTO SAVANA_ORDERS
  2 VALUES
  3 (1002, 102, TO_DATE('21-09-2026','DD-MM-YYYY'), 1999, 'Shipped');

1 row created.

SQL>
SQL> INSERT INTO SAVANA_ORDERS
  2 VALUES
  3 (1003, 103, TO_DATE('20-09-2026','DD-MM-YYYY'), 1598, 'Delivered');

1 row created.

SQL>
SQL> COMMIT;

Commit complete.

```

5.Order_Details Data Insertion

```
SQL> SELECT PRODUCT_ID, PRODUCT_NAME  
2 FROM SHOP_PRODUCT;
```

```
PRODUCT_ID  
-----  
PRODUCT_NAME  
-----
```

```
101  
Wireless Headphones
```

```
105  
Mixer Grinder
```

```
106  
Bluetooth Speaker
```

```
PRODUCT_ID  
-----  
PRODUCT_NAME  
-----
```

```
107  
Casual Shirt
```

```
108  
Face Wash
```

```
102  
Smart Watch
```

```
PRODUCT_ID  
-----  
PRODUCT_NAME  
-----
```

```
103  
Cotton Shirt
```

```
104  
Denim Jeans
```

```
109  
Handbag
```

```
PRODUCT_ID  
-----  
PRODUCT_NAME  
-----
```

```
110  
Electric Kettle
```

```
10 rows selected.
```

6.Order Modification Operations

6.1 Modify Order Amount

```
SQL> UPDATE SAVANA_ORDERS  
2 SET TOTAL_AMOUNT = 26998  
3 WHERE ORDER_ID = 1001;
```

```
1 row updated.
```

6.2 Modify Order Date

```
SQL> UPDATE SAVANA_ORDERS
  2 SET ORDER_DATE = TO_DATE('22-09-2026', 'DD-MM-YYYY')
  3 WHERE ORDER_ID = 1001;

1 row updated.
```

7.Order_Details Modification

7.1 Modify Quantity

```
SQL> UPDATE SAVANA_ORDER_DETAILS
  2 SET QUANTITY = 2,
  3     SUBTOTAL = 15999 * 2
  4 WHERE ORDER_DETAIL_ID = 1;

0 rows updated.
```

7.2 Display Modified Details

```
SQL> SELECT *
  2 FROM SAVANA_ORDER_DETAILS
  3 WHERE ORDER_DETAIL_ID = 1;

no rows selected
```

8. Customer Order History Report

```

SQL> SELECT
  2     O.ORDER_ID,
  3     O.CUSTOMER_ID,
  4     O.ORDER_DATE,
  5     D.PRODUCT_ID,
  6     D.QUANTITY,
  7     D.UNIT_PRICE,
  8     D.SUBTOTAL,
  9     O.TOTAL_AMOUNT,
10     O.ORDER_STATUS
11 FROM SAVANA_ORDERS O
12 JOIN SAVANA_ORDER_DETAILS D
13 ON O.ORDER_ID = D.ORDER_ID
14 ORDER BY O.ORDER_DATE DESC;

no rows selected

```

9.Customer-Wise Order Summary

```

SQL> SELECT
  2     CUSTOMER_ID,
  3     COUNT(ORDER_ID) AS TOTAL_ORDERS,
  4     SUM(TOTAL_AMOUNT) AS TOTAL_PURCHASE_AMOUNT
  5 FROM SAVANA_ORDERS
  6 GROUP BY CUSTOMER_ID
  7 ORDER BY CUSTOMER_ID;

```

CUSTOMER_ID	TOTAL_ORDERS	TOTAL_PURCHASE_AMOUNT
-----	-----	-----
101	1	26998
102	1	1999
103	1	1598